

Research Proposal: Structural Dynamics and Asymmetric Pass-Through of Fuel Tax Interventions

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1 Research Summary

This project investigates the impact of temporary fiscal policy on retail fuel markets in the form of excise tax cuts in a special setting characterized by cycling prices. The natural experiment provided by the post-COVID inflation-relief excise tax cut in Australia presents an opportunity to study this phenomenon, which is currently significant in light of the recent global oil price shock.

Oil prices are well-known to follow an asymmetric cycling pattern in Australia, in which industry competitors alternate in undercutting prices until a certain price floor, then a discrete jump-back to a higher price signals the start of another cycle. By employing a Markov Switching Model (Noel, 2007), the study classifies stations into latent pricing regimes of “undercutting,” “stable,” and “relenting” to evaluate whether a major tax cut alters structural competition parameters. Using a Regression Discontinuity Design (RDD) in Time, we measure the extent to which the tax cut, as an exogenous cost-shift, was efficiently passed-through to consumers. Specifically, we ask if the answers to such questions differ across markets endowed with distinct cycling patterns, quantifying the asymmetric effect of regulatory “jawboning” (i.e., regulatory threats to firms in non-legal forms) during the restoration phase.

2 Contextual Background

The 2022 Australian fuel excise intervention serves as a unique natural experiment to study the transmission of fiscal shocks. As documented by the ACCC, Australian petrol pricing is defined by Edgeworth Cycles—a “sawtooth” pattern of sharp hikes and slow decays. Recent international evidence suggests that such “inflation relief” tax cuts often result in incomplete pass-through (Noel, Roach, and Hill, 2024). This research examines whether these cycles, which vary from the deterministic 24-hour cycle in Perth to stochastic coordination in Sydney (Byrne and de Roos, 2019), facilitate or obscure the pass-through of tax changes to consumers.

3 Research Questions

- **RQ1:** Does a large exogenous cost shock induce a structural break in the latent state transition probabilities of Edgeworth cycles (Noel, 2007)?
- **RQ2:** To what extent does the magnitude of pass-through differ between deterministic, stochastic, and non-cycling regimes?

- **RQ3:** What is the relationship between spatial competition density, local market power, and the speed of price adjustment (Deltas, 2008)?
- **RQ4:** Is there a persistent “Rockets and Feathers” effect (Deltas, 2008) during tax restoration, and can it be statistically mitigated by ACCC regulatory signaling (jawboning)?

4 Data Requirements and Sources

Table 1: Data Streams and Primary Sources

Variable	Description	Primary Source
Retail Prices (WA)	Daily prices governed by the “24-hour rule.”	FuelWatch (Data.WA)
Retail Prices (NSW)	Real-time station-level reporting (>2,500 stations).	FuelCheck (Data.NSW)
Wholesale Cost	Daily Terminal Gate Prices (TGP).	AIP / FuelWatch
Tax Policy	Excise rates (44.2 vs 22.1 cpl).	Australian Treasury
Regulatory Signal	ACCC media releases for jawboning intensity.	ACCC Newsroom

References

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